SDLC PROJECT REPORT

Toy Store E-Commerce

Sales Analytics & Business Intelligence Dashboard

SDLC Project Document

Project Duration: 5th June 2026 – 5th August 2026

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Abstract

This report documents the design, development, and analytical output of a Power BI business-intelligence solution built for a fictional e-commerce toy retailer, The Toy Store. The project was executed by an eleven-member analytics team over a two-month engagement (5 June - 5 August 2026), following a structured Software/Analytics Development Life Cycle (SDLC) spanning requirement gathering, data staging, data-quality assurance, semantic modelling, dashboard construction, and validation.

Approximately 1.7 million rows across six relational tables: orders, order items, order-item refunds, products, website sessions, and website pageviews. These tables were uploaded to Microsoft SQL Server and validated in SQL Server Management Studio (SSMS) before being modelled in Power BI Desktop. A star-style semantic model with a dedicated Date table and a 42-measure DAX catalogue supports seven interactive report pages: Executive Summary, Strategic Takeaways, Product Analysis, Customer Behaviour, Transaction Analytics, Inventory Analysis, and Website Session, all bound together by a consistent set of 19 slicers and a page-navigator control.

Over the analysed period (2012-2015) the business generated $1.94M in revenue and $1.22M in profit across 32K orders and 31.70K customers, at an average order value of $59.99. The report walks through the data-preparation methodology, the entity-relationship model, the complete measure catalogue with representative DAX formulas, a page-by-page tour of the dashboard with embedded screenshots, and a synthesis of the strategic findings including the dominance of a single hero product, a Google-search-led acquisition channel, and a thin gold-tier customer base together with recommendations for inventory, marketing, and retention strategy.

1. Introduction

1.1 Project Background

E-commerce operations produce large volumes of transactional and behavioural data from diverse sources such as sales orders, refunds, product inventories, and website clickstream activities. In its raw state, this data provides limited business value. The Toy Store project was undertaken as a comprehensive analytics exercise to collect, clean, validate, and transform these operational datasets into an interactive Power BI dashboard ecosystem. The solution enables non-technical stakeholders including senior management, merchandising teams, and marketing departments to monitor performance, identify trends, and support data-driven decision-making at both strategic and operational levels.

1.2 Objectives

- Give top management a single-glance view of revenue, profit, order, and customer KPIs.
- Break down performance by product to support data-driven merchandising decisions.
- Understand customer segments like new vs. repeat, at-risk vs. active, based on device type and their spending behaviour.
- Provide transaction-level and website-session-level detail for operational analysis.
- Support self-service exploration through a consistent slicer and navigation experience on every page.

1.3 Scope of Work

The scope covers the full analytics pipeline: sourcing and staging six raw tables in SQL Server, performing systematic data-quality checks (nulls, duplicates, statistical outliers, and business-rule validation), building a semantic model with a dedicated date dimension, authoring a 36-measure DAX catalogue organised into four analytical themes, and delivering seven Power BI report pages covering executive, strategic, product, customer, transaction, inventory, and web-traffic analysis.

1.4 Tools & Technology Stack

Layer	Tool	Purpose
Data Staging & QA	Microsoft SQL Server / SSMS	Loading raw CSV exports, null/duplicate/outlier checks, business-rule validation
Data Modelling	Power BI Desktop (Power Query, Tabular Model)	Relationships, star schema, Date table, calculated columns
Analytics Layer	DAX (Data Analysis Expressions)	42 measures across Revenue & Profit, Orders & Customers, Product & Returns, Website & Sessions
Presentation Layer	Power BI report canvas	7 report pages, 30 visuals, 19 slicers, page-navigator buttons

2. Project Methodology & Development Life Cycle

The project followed a linear, checkpoint-driven analytics development life cycle, moving from raw operational exports through to a validated, business-ready dashboard suite. Each stage produced an artefact that fed the next: cleaned tables from SQL Server became the source for the Power BI semantic model, which in turn became the foundation for the report pages and measure catalogue.

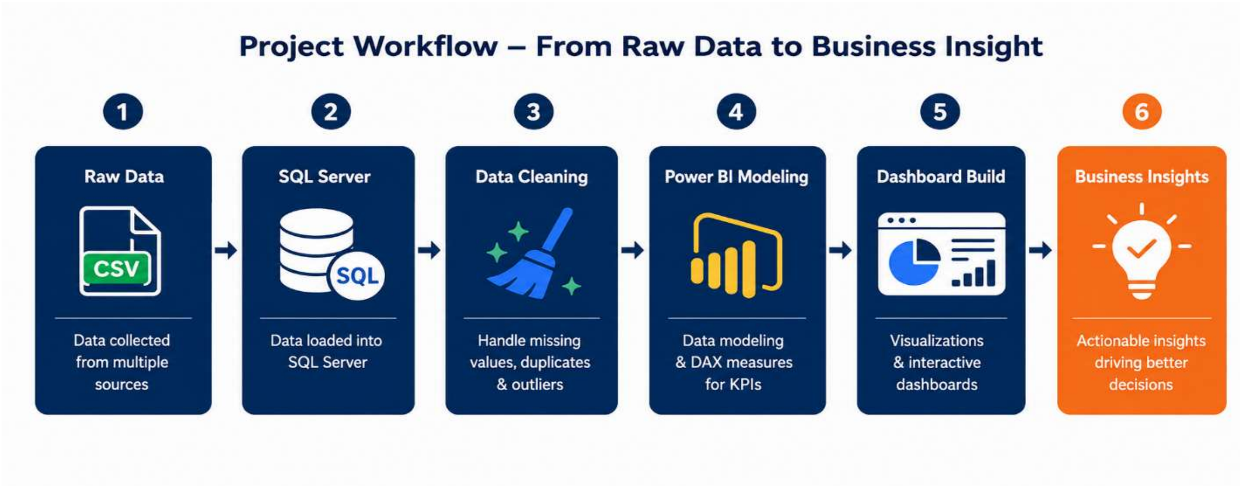

Figure 2.1 — End-to-end project workflow, from raw CSV exports to executive-ready business insight.

2.1 Life-Cycle Stages

1. **Requirement Gathering:** Objectives, KPI list, and stakeholder page requirements were agreed with the analytics team (11 members) at project kickoff.
2. **Data Acquisition & Staging**: Six raw CSV exports (orders, order items, order-item refunds, products, website sessions, website pageviews) were loaded into SQL Server.
3. **Data Cleaning & Validation:** SSMS was used to run systematic null, duplicate, outlier, and business-logic checks across all six tables (~17 lakh / 1.7 million rows).
4. **Data Modelling:** tables were imported into Power BI Desktop, relationships were established (Entity-Relationship Diagram), and a dedicated Date table was added to drive time intelligence.
5. **DAX Measure Development**: 42 measures were authored in a single "All Measures" folder, grouped into four analytical themes.
6. **Dashboard Creation**: Seven report pages were built with 30 total visuals and 19 cross-page slicers.
7. **Validation & Sign-off:** Visuals were checked against source aggregates, and the strategic takeaways captured on each page were reviewed by the team before final delivery on 5th August June 2026.

2.2 Team Structure

The project was delivered by an eleven-member analytics team: Arpita Banik, Debanjan Sengupta, Puja Kumari, Rakhi Biswas, Riya Majhi, Rohit Ghosh, Sayan Ghosh, Siuli Bhowmick, Subham Hazra, Tanushree Seal, and Tushar Mishra, working collaboratively across data engineering, modelling, and dashboard design responsibilities, guided by Mr. Somujit Biswas and Mr. Amit Chaterjee.

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3. Data Architecture & Source Tables

Six relational tables sourced from the toy store's transactional and web-analytics systems form the backbone of the model. Together they span roughly 1.7 million rows across a March 2012 - March 2015 operating window.

3.1 Dataset Summary

Table	Rows	Date Range	Notes
orders	32,313	Mar 2012 – Mar 2015	No nulls · no duplicates · COGS always < price
order_items	40,025	Mar 2012 – Mar 2015	No nulls · no duplicates
order_item_refunds	1,731	Apr 2012 – Apr 2015	No nulls · no duplicates
products	4	Mar 2012 – Feb 2014	Reference table — verified complete
website_pageviews	1,188,124	Mar 2012 – Mar 2015	No nulls · no duplicates
website_sessions	472,871	Mar 2012 – Mar 2015	utm_campaign / utm_content blank for 83,328 "Direct" sessions

3.2 Table Roles

- orders — one row per completed order; carries user_id, items_purchased, Customer_Category, and order-level price/COGS totals.
- order_items — line-item detail beneath each order, including per-item price, COGS, and a flag for the primary cart item versus cross-sold additions.
- order_item_refunds — refund events linked back to order items, used to compute return rate and revenue-at-risk.
- products — the four-SKU product catalogue (The Original Mr. Fuzzy, The Forever Love Bear, The Birthday Sugar Panda, The Hudson River Mini Bear).
- website_sessions — one row per browsing session, carrying utm_source / utm_campaign, device_type, and a repeat-session flag used for new-vs-returning analysis.
- website_pageviews — page-level clickstream detail linked to website_sessions, used for funnel and top-pages analysis.
- Date Table — a standalone calendar dimension (Year, Month, Month Year) that drives all time-intelligence measures and the Year/Month slicers.

4. Data Preparation & Quality Assurance (SSMS)

Before any modelling took place in Power BI, all six raw CSV exports were loaded into SQL Server and worked through in SQL Server Management Studio. Four categories of checks were performed across the full ~17-lakh-row dataset: null-value review, duplicate-row detection, statistical outlier screening, and business-rule validation.

6	17L	0	2
Tables Loaded	Total Rows	Duplicate Rows Found	Columns with Expected Nulls

4.1 Null-Value Review

Every table was scanned column-by-column for NULL values using conditional aggregation. Only the utm_campaign and utm_content columns in website_sessions returned nulls, and only for the 83,328 sessions correctly attributed to “Direct” traffic — an expected condition rather than a data-quality defect, since direct visits carry no campaign tagging.

```
-- Null-value review: run once per table, one column pair at a time
SELECT
    COUNT(*)                                            AS total_rows,
    SUM(CASE WHEN order_id       IS NULL THEN 1 ELSE 0 END) AS null_order_id,
    SUM(CASE WHEN user_id        IS NULL THEN 1 ELSE 0 END) AS null_user_id,
    SUM(CASE WHEN created_at     IS NULL THEN 1 ELSE 0 END) AS null_created_at,
    SUM(CASE WHEN price_usd      IS NULL THEN 1 ELSE 0 END) AS null_price_usd,
    SUM(CASE WHEN cogs_usd       IS NULL THEN 1 ELSE 0 END) AS null_cogs_usd
FROM dbo.orders;

-- Expected-null case: website_sessions campaign fields
SELECT COUNT(*) AS direct_sessions_with_blank_campaign
FROM dbo.website_sessions
WHERE utm_source = 'Direct'
  AND (utm_campaign IS NULL OR utm_content IS NULL);
```

4.2 Duplicate-Row Detection

A full-row duplicate scan was executed across all six tables using GROUP BY / HAVING on the natural and surrogate keys. No duplicate rows were found in any table.

```
-- Duplicate check pattern, applied to each of the 6 tables
SELECT order_item_id, COUNT(*) AS occurrences
FROM dbo.order_items
GROUP BY order_item_id
HAVING COUNT(*) > 1;

-- Full-row duplicate variant (catches accidental re-imports)
SELECT order_id, user_id, created_at, items_purchased, price_usd, cogs_usd,
       COUNT(*) AS occurrences
FROM dbo.orders
```

```
GROUP BY order_id, user_id, created_at, items_purchased, price_usd, cogs_usd
HAVING COUNT(*) > 1;
```

4.3 Statistical Outlier Check (Z-Score)

A Z-score based check was run on price_usd, cogs_usd, refund_amount_usd, and items_purchased across all relevant tables to flag statistically unusual values (|Z| > 3). The check surfaced a large share of price, COGS, and items-purchased values as flagged, but on inspection these reflected legitimate variation across only four SKUs and 1-2 item orders, not data-entry errors. Because the product catalogue is small, price and COGS naturally cluster into a handful of discrete values, which inflates the apparent Z-score dispersion without indicating bad data.

```
-- Z-score outlier check on order_items.price_usd
with Stats as(
select order_item_id, price_usd,
AVG(price_usd) over() as avg_price,
STDEV(price_usd) over() as std_dev
from order_items)

select order_item_id, price_usd,
((price_usd-avg_peice)/std_dev) as z_score
from Stats
where ABS(z_score)>3
```

4.4 Business-Logic Validation

Finally, the data was validated against expected business rules: COGS must always be lower than price, no monetary field may be negative or zero, and all dates must fall within the valid operating window. All six tables passed these checks with zero violations.

```
-- Business-rule validation: COGS must always be below price
SELECT order_item_id, price_usd, cogs_usd
FROM dbo.order_items
WHERE cogs_usd >= price_usd;

-- No negative or zero monetary values
SELECT 'orders' AS tbl, order_id AS id FROM dbo.orders
  WHERE price_usd <= 0 OR cogs_usd <= 0
UNION ALL
SELECT 'order_item_refunds', order_item_refund_id FROM dbo.order_item_refunds
  WHERE refund_amount_usd <= 0;

-- Valid date range check
SELECT * FROM dbo.orders
WHERE created_at < '2012-01-01' OR created_at > '2015-12-31';
```

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5. Data Model & Entity-Relationship Diagram

The semantic model follows a star-style pattern: two fact-like tables (orders and order_items, with order_item_refunds as a refund sub-fact) connect outward to the products dimension and, via user/session identifiers and the shared Date table, to the website_sessions and website_pageviews web-analytics tables. All 42 DAX measures live in a dedicated, columnless “All Measures” table so that report authors can find every calculation in one place rather than hunting through source tables.

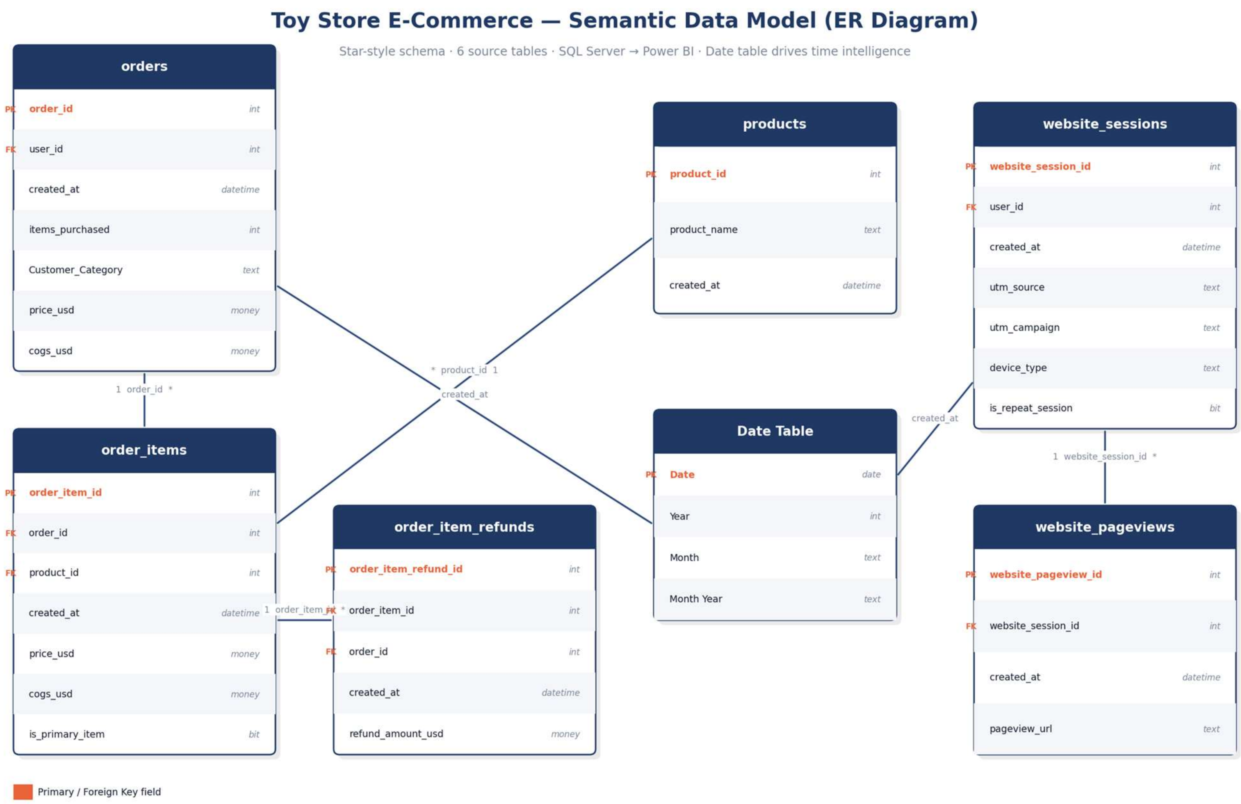

Figure 5.1 — Entity-relationship diagram of the Toy Store semantic model (6 source tables + Date table).

5.1 Key Relationships

From	To	Cardinality	Join Key
orders	order_items	1 : *	order_id
order_items	order_item_refunds	1 : *	order_item_id
order_items	products	* : 1	product_id
website_sessions	website_pageviews	1 : *	website_session_id
orders / website_sessions	Date Table	* : 1	created_at → Date

5.2 Modelling Notes

- The Date table is marked as the model's official date table, which enables native DAX time-intelligence functions (e.g. rolling averages, year-over-year comparisons) used in the Transaction Analytics and Inventory Analysis pages.
- order_items separates the primary cart item from cross-sold additions via an is_primary_item-style flag, which underlies the “Cart Item and Cross-Sold Addition” visual on the Product Analysis page.
- Both orders and website_sessions relate independently to the Date table, allowing revenue and traffic trends to be sliced on a common timeline even though they are not directly related to one another.

5.3 Calculated Table Creation: Date Table

The primary objective of creating this calculated Date Table (also known as a Calendar Table) is to serve as the single, continuous time intelligence foundation for your entire Power BI data model.

```
Date Table =
ADDCOLUMNS(
CALENDAR(
MIN(Orders[created_at]),
MAX(Orders[created_at])
),
"Year",YEAR([Date]),
"Month No",MONTH([Date]),
"Month",FORMAT([Date],"MMMM"),
"Month Year",FORMAT([Date],"MMM YYYY"),
"Quarter","Q"&FORMAT([Date],"Q"),
"Week",WEEKNUM([Date]),
"Day",DAY([Date]),
"Day Name",FORMAT([Date],"dddd")
)
```

Figure 5.3 — DAX for Calculated Table: Date Table Creation

6. DAX Measure Catalogue

Thirty-six DAX measures live in the model's "All Measures" folder, grouped into seven analytical themes that mirror the report's page structure. The formulas below are presented in standard DAX syntax and reflect the calculation logic implied by each measure's name, its position in the model, and its behaviour across the report's slicers.

6.1 Executive Overview:

```
Total Orders =
DISTINCTCOUNT(Orders[order_id])
```

```
Total Customers =
DISTINCTCOUNT(orders[user_id])
```

```
Total Revenue =
CALCULATE(
    SUM(Orders[price_usd]),
    TREATAS(VALUES(products[product_id]), Orders[primary_product_id])
)
```

```
Total Profit =
[Total Revenue]-[Total COGS]
```

```
Aov = DIVIDE(
    SUM(orders[price_usd]),
    DISTINCTCOUNT(orders[order_id]),
    0
)
```

6.2 Strategic Takeaways

```
Total Revenue =
CALCULATE(
    SUM(Orders[price_usd]),
    TREATAS(VALUES(products[product_id]), Orders[primary_product_id])
)
```

```
Aov = DIVIDE(
    SUM(orders[price_usd]),
    DISTINCTCOUNT(orders[order_id]),
    0
)
```

6.3 Product Analysis

```
cross-sold-addition = CALCULATE(COUNT(order_items[order_id]),'order_items'[is_primary_item] = 0)
```

```
main-cart-item = CALCULATE(COUNT('order_items'[order_id]),'order_items'[is_primary_item] = 1)
```

```
Revenue At Risk =
CALCULATE(
    SUM('Orders'[price_usd]),
    TREATAS(VALUES('products'[product_id]), 'Orders'[primary_product_id]),
    FILTER(
        VALUES('Orders'[user_id]),
        [Days Since Last Purchase] >= 90
    )
)
```

```
Total Profit =
[Total Revenue]-[Total COGS]
```

```
Total Revenue =
CALCULATE(
    SUM(Orders[price_usd]),
    TREATAS(VALUES(products[product_id]), Orders[primary_product_id])
)
```

```
Total COGS =
CALCULATE(
    SUM(Orders[cogs_usd]),
    TREATAS(VALUES(products[product_id]), Orders[primary_product_id])
)
```

```
Total_Refunds_Issued = COUNT(order_item_refunds[order_item_refund_id])
```

```
Product_Return_Rate_% =
DIVIDE(
    CALCULATE(
        COUNT(order_item_refunds[order_item_refund_id]),
        TREATAS(VALUES(products[product_id]), order_items[product_id])
    ),
    CALCULATE(
        COUNT(order_items[order_item_id]),
        TREATAS(VALUES(products[product_id]), order_items[product_id])
    ),
    0
)
```

6.4 Customer Behaviour

```
Unique Customers = DISTINCTCOUNT(orders[user_id])
```

```
Repeat_Audience_Share_% =
        DIVIDE(
        CALCULATE(COUNT(website_sessions[website_session_id]), website_sessions[is_repeat_session] = 1),
        COUNT(website_sessions[website_session_id]),
        0
        )
```

```
Total At-Risk Customers = CALCULATE(DISTINCTCOUNT('Orders'[user_id]), 'Orders'[Customer Status] = "At Risk of Churn")
```

```
new customer traffic = CALCULATE([traffic source],website_sessions[is_repeat_session] = 0)
```

```
repeated customer traffic = CALCULATE([traffic source],website_sessions[is_repeat_session] = 1)
```

```
Product_Return_Rate_% =
DIVIDE(
    CALCULATE(
        COUNT(order_item_refunds[order_item_refund_id]),
        TREATAS(VALUES(products[product_id]), order_items[product_id])
    ),
    CALCULATE(
        COUNT(order_items[order_item_id]),
        TREATAS(VALUES(products[product_id]), order_items[product_id])
    ),
    0
)
```

```
Segment Average Order Value AOV =
VAR Segment_Completed_orders= DISTINCTCOUNT('Orders'[order_id])
VAR Segment_Total_Revenue = SUM('Orders'[price_usd])

RETURN
DIVIDE(Segment_Total_Revenue, Segment_Completed_orders, 0)
```

```
average order per customer = var orders = COUNT(Orders[order_id])
var customers = DISTINCTCOUNT(Orders[user_id])
RETURN
DIVIDE(orders,customers,0)
```

6.5 Transaction Analysis

```
Total Revenue =
CALCULATE(
    SUM(Orders[price_usd]),
    TREATAS(VALUES(products[product_id]), Orders[primary_product_id])
)
```

```
Total Orders =
DISTINCTCOUNT(Orders[order_id])
```

```
Total Profit =
[Total Revenue]-[Total COGS]
```

```
Avg_Basket_Size = AVERAGE(Orders[items_purchased])
```

```
Total Items Purchased =
SUM(Orders[items_purchased])
```

```
Aov = DIVIDE(
    SUM(orders[price_usd]),
    DISTINCTCOUNT(orders[order_id]),
    0
)
```

```
Sales_Volume_3M_Rolling_Avg =
AVERAGEX(
    DATESINPERIOD('Date Table'[Date], MAX('Date Table'[Date]), -3, MONTH),
    COUNT(Orders[order_id])
)
```

```
Total COGS =
CALCULATE(
    SUM(Orders[cogs_usd]),
    TREATAS(VALUES(products[product_id]), Orders[primary_product_id])
)
```

6.6 Inventory Analysis

```
Total Units Sold =
COUNT(order_items[order_item_id])
```

```
Avg Profit Margin Per Transaction % =
AVERAGEX(
    orders,
    DIVIDE(
        orders[price_usd] - Orders[cogs_usd],
        orders[price_usd],
        0
    )
)
```

```
Revenue At Risk =
CALCULATE(
    SUM('Orders'[price_usd]),
    TREATAS(VALUES('products'[product_id]), 'Orders'[primary_product_id]),
    FILTER(
        VALUES('Orders'[user_id]),
        [Days Since Last Purchase] >= 90
    )
)
```

```
Total Gross Revenue = SUM('order_items'[price_usd])
```

```
Total Profit =
[Total Revenue]-[Total COGS]
```

```
3-Month Rolling Avg Sales Volume =
AVERAGEX(
    DATESINPERIOD(
        'Date Table'[Date],
        MAX('Date Table'[Date]),
        -3,
        MONTH
    ),
    CALCULATE([Total Sales Volume])
)
```

```
Monthly Sales Volume = CALCULATE([Total Sales Volume])
```

```
Total Units Sold = COUNT(order_items[order_item_id])
```

```
Product Sales = SUM(order_items[price_usd])
```

6.7 Website Session

```
Total Sessions = DISTINCTCOUNT('website_sessions'[website_session_id])
```

```
Total_page_views = COUNTROWS(website_pageviews)
```

```
Avg Pages Per Session =
DIVIDE([Total_page_views], [Total Sessions])
```

```
Visitor Conversion Rate =
DIVIDE([Total Orders], [Total Sessions])
```

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7. Report Pages

The dashboard is organised into seven pages, each addressing a distinct analytical lens, and each reachable from a consistent page-navigator control at the top of the canvas. This section presents every page's purpose, its embedded screenshot, and the strategic takeaways the team captured while building it.

7.1 Executive Summary

The landing page of the report presents company-wide KPI cards : total revenue, profit, orders, customers, and average order value for the currently selected period, alongside the dashboard cover, project credits, a revenue/profit/refund comparison chart, and a profit-contribution-by-year donut chart.

Figure 7.1 — Executive Summary page of the Toy Store dashboard.

Key Insights

- $1.22M profit on $1.94M revenue implies a low relative cost of goods sold or premium pricing, this a very strong margin for e-commerce retail.
- Refunds sit near $0.1M against $1.94M in revenue, roughly 5%, returns are not a material risk as of today.
- 2014 alone generated 63.69% of all profit, with 2013 a distant second at 18.92%, showing heavy year-over-year concentration.

7.2 Strategic Takeaways

An continuation view that keeps the core KPI cards visible while adding a monthly revenue trend line, a revenue-by-traffic-source breakdown, a monthly average-order-value bar chart, and a revenue-contribution-by-year donut.

Figure 7.2 — Strategic Takeaways page of the Toy Store dashboard.

Key Insights

- Monthly revenue peaks significantly in June ($219.95) and November ($225.95), and drops to its lowest points in March ($45.99) and September ($49.99), finishing the year at $79.98 in December.
- Google search (gsearch) drives $491.89 in revenue, more than bsearch ($275.94) and Direct ($59.99) combined, confirming it as the primary growth channel.
- Monthly average order value peaks in August ($95.98) even though revenue that month is mid-range, meaning customers bought less often but spent more per order.
- Revenue contribution in 2012 (18.84%), 2013 (12.08%), and 2015 (6.04%) trail well behind 2014 (63.04%), this is the same concentration pattern seen in the Executive profit-by-year view.

7.3 Product Analysis

Focused on SKU-level performance: which products drive revenue and profit, which are returned or refunded most, their share of cost of goods sold, and cross-sell behaviour at the cart level.

Figure 7.3 — Product Analysis page of the Toy Store dashboard.

Key Insights

- The Original Mr. Fuzzy dominates overall revenue, profit, and potential revenue at risk (above $1M); the remaining three products generate significantly lower financial volume, with The Hudson River Mini Bear contributing minimal revenue and profit.
- The vast majority of production/inventory cost belongs to the top-selling product, roughly matching its share of overall sales volume.
- Refunds are evenly split across all four products (25% each), total refund count is uniform regardless of sales-volume differences between products.
- There is very high cross-sell conversion: for every 2 primary items added to a cart, 1 additional recommended item is successfully added, suggesting an opportunity to attach cross-sells to lower-performing items like The Hudson River Mini Bear.
- More than 90% of total returns come from the low-performing products, despite their small share of sales volume.

7.4 Customer Behaviour

Segments the customer base by new-vs-repeat status, at-risk-vs-active status, and device type, presenting basket size and average order value by segment alongside traffic-source and loyalty-tier breakdowns.

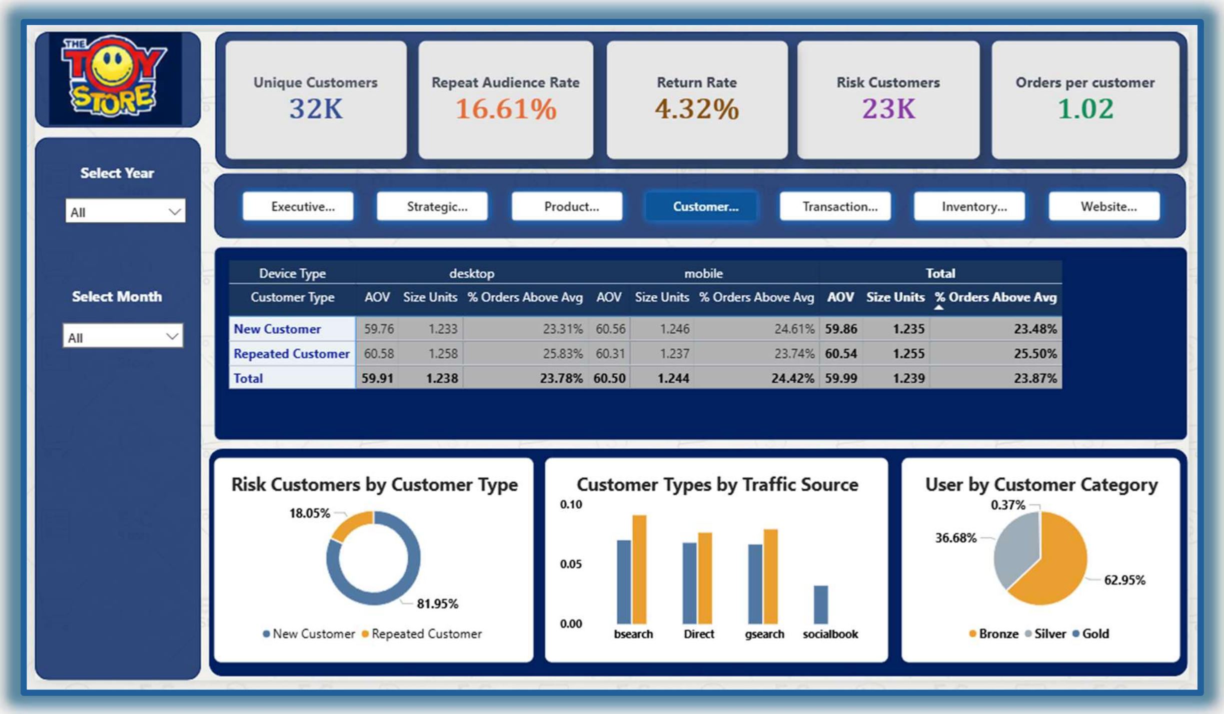

Figure 7.4 — Customer Behaviour page of the Toy Store dashboard.

Key Insights

- Average order value is extremely consistent across customer segments: new vs. repeat, desktop vs. mobile all sit within a $1-2 band of $59.99-$60.58.
- First-time buyers present significantly higher retention or transaction risk than returning buyers (81.95% of at-risk customers are new customers).
- bsearch, Direct, and gsearch attract a higher proportion of repeated customers than new customers, while socialbook drives new customers almost exclusively, with negligible representation from returning customers.
- More than 99% of the entire user base sits in the entry (Bronze) or mid (Silver) loyalty tier; the transition from Bronze to Silver is strong, but Silver-to-Gold conversion is very weak, suggesting the Gold-tier threshold or benefits merit review.

7.5 Transaction Analytics

Order-level operational detail: an average-order-value gauge against a target threshold, a 3-month rolling average of order/sales volume by year and month, total COGS vs. total profit, order composition by items purchased, and revenue by product.

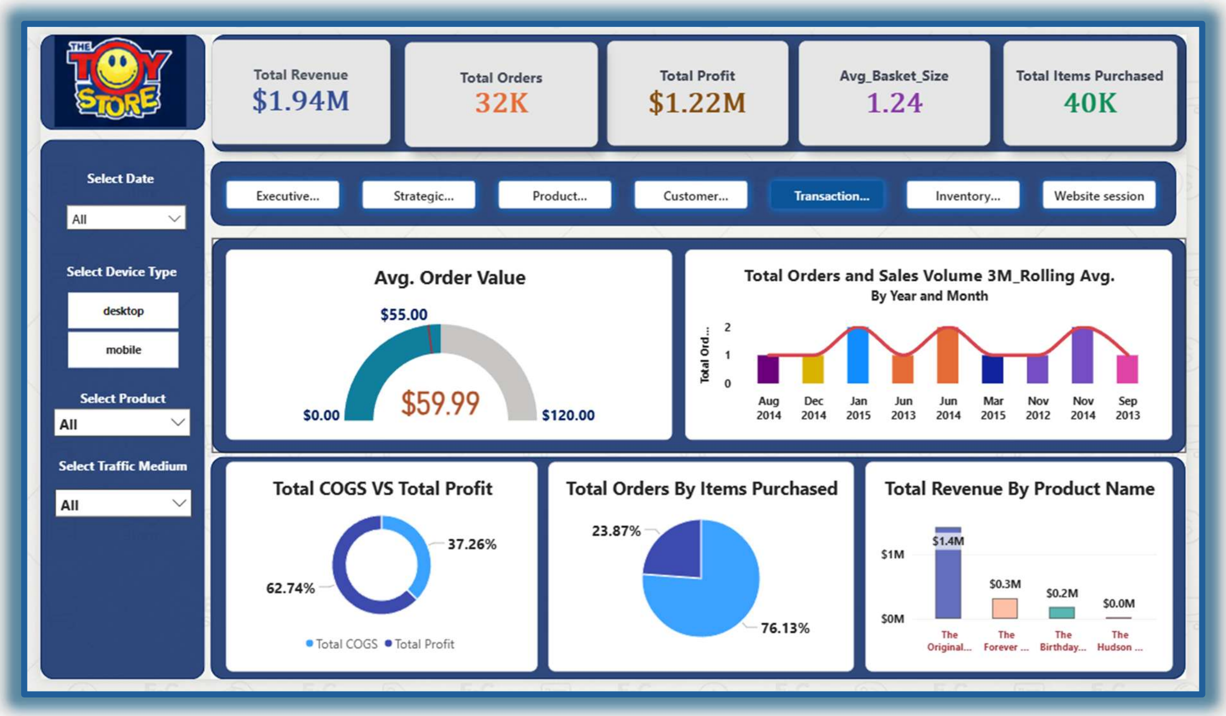

Figure 7.5 — Transaction Analytics page of the Toy Store dashboard.

Key Insights

- The current AOV sits at $59.99 against a target threshold marked around $55.00, current average spending beats the baseline target by roughly $5.00.
- Order volume reaches its highest points (2 orders in the sampled view) in January 2015, June 2014, and November 2014, with a recurring wave pattern highlighting mid-year and year-end demand spikes.
- The business retains over 62 cents of gross profit for every dollar generated, indicating strong margins with comfortable room for paid marketing or promotional discounting.
- Over three-quarters of all orders contain a single item, The Original Mr. Fuzzy, while only about a quarter of orders include the other three products.
- The Original Mr. Fuzzy generates $1.4M, dominating overall business revenue; The Forever Love Bear generates $0.3M, The Birthday Sugar Panda $0.2M, and The Hudson River Mini Bear a negligible contribution.

7.6 Inventory Analysis

Stock-relevant metrics: units sold and sales-velocity trend by year, a sales funnel of total units sold by product, an inventory-risk view weighting exposure by SKU, and new-vs-returning session share by product.

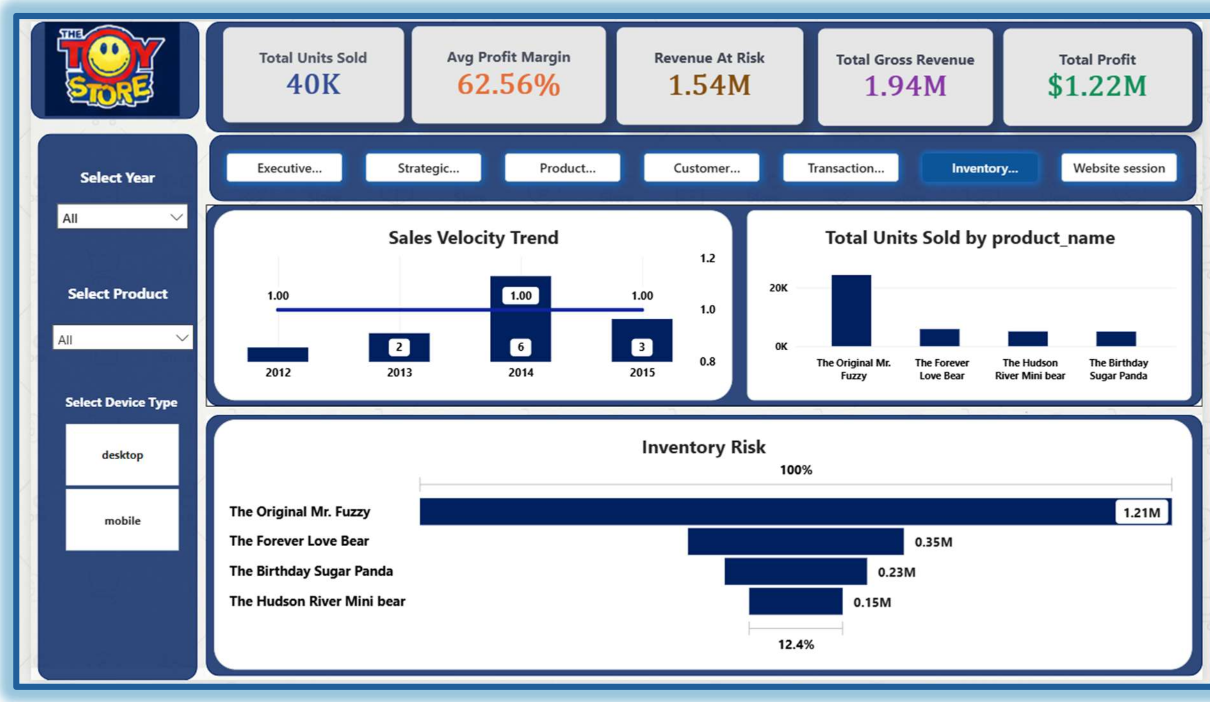

Figure 7.6 — Inventory Analysis page of the Toy Store dashboard.

Key Insights

- Sales velocity peaked in 2014 at a score of 6, up from 2 in 2013, before dropping by half to 3 in 2015, still above 2012-2013 levels, while the benchmark trendline holds steady at 1.00 throughout.
- The Original Mr. Fuzzy dominates unit sales by a massive margin (exceeding 20,000 units), while the other three products cluster tightly at roughly 2,000-4,000 units each.
- The Original Mr. Fuzzy represents the vast majority of inventory risk value (1.21M), making overall inventory health almost entirely tied to a single SKU; secondary SKUs range down to 0.15M, only 12.4% of top-tier exposure.
- The Original Mr. Fuzzy also drives 1.21M sessions, matching its dominance in sales and inventory footprint, while the other three products capture far less top-of-funnel traffic.

7.7 Website Session

Top-of-funnel web analytics covering total sessions, page views, average pages per session, visitor conversion rate, and revenue per session, alongside traffic-source conversion, top-viewed pages, monthly session trend, and new-vs-returning sessions by year.

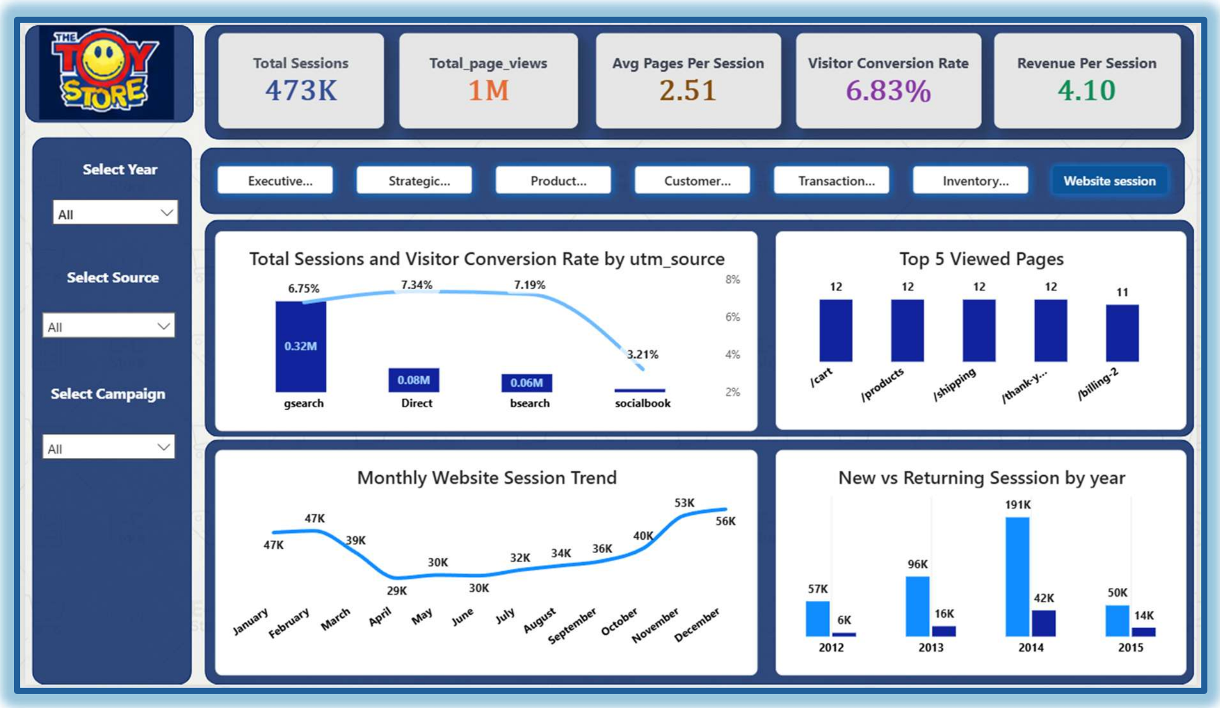

Figure 7.7 — Website Session page of the Toy Store dashboard.

Key Insights

- gsearch dominates traffic volume at 0.32M sessions with a strong 6.75% conversion rate; Direct traffic generates 0.08M sessions with the highest conversion rate at 7.34%, and bsearch delivers 0.06M sessions at 7.19%. Socialbook drives minimal traffic with the lowest conversion rate at 3.21% which is less than half of the search/direct channels.
- cart, products, shipping, and thank-you pages share equal top pageviews (12 each), followed by billing (11), indicating high-intent traffic progressing evenly through the core checkout path.
- Website traffic begins strong in January-February (47K), dips to its lowest point in April (29K), then recovers steadily through the year to peak during the November-December holiday season (53K-56K).
- Total traffic peaked massively in 2014 at 191K new and 42K returning sessions, before dropping substantially in 2015 to 50K new and 14K returning sessions; across all years, new-user sessions make up roughly 75-90% of total web traffic, highlighting a heavily acquisition-driven funnel.

8. Filters, Slicers & Interactivity

Interactivity is delivered through 19 slicers distributed across the seven pages, plus a persistent page-navigator control that lets users move between any of the seven views with a single click from anywhere in the report.

Slicer / Filter	Applied On	Purpose
Date (Month / Year)	Executive Summary, Strategic Takeaways, Product Analysis, Customer Behaviour, Inventory Analysis, Website Session	Restricts all visuals on the page to a selected year and/or month
Device Type	Transaction Analytics, Inventory Analysis	Splits KPIs and visuals by desktop vs. mobile
Product	Transaction Analytics, Inventory Analysis	Filters transaction and inventory visuals to a single SKU
Traffic Source / Medium	Strategic Takeaways, Customer Behaviour, Transaction Analytics, Website Session	Filters by utm_source / utm_medium (gsearch, bsearch, Direct, socialbook)
Campaign	Website Session	Filters by utm_campaign
Page Navigator	All 7 pages	Click-through buttons for one-click navigation between report pages

Because the Date, Device Type, Product, and Traffic Source filters are repeated with identical behaviour across multiple pages, users can carry a mental model of “what this control does” from one page to the next, reducing the learning curve for a seven-page report.

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9. Key Findings & Business Recommendations

9.1 Consolidated Findings

- Single-product concentration risk. The Original Mr. Fuzzy accounts for the large majority of revenue, profit, units sold, sessions, and inventory-risk exposure. This is both the business's greatest strength (a proven bestseller) and its greatest structural risk (a single point of failure).
- Strong, stable margins. The business retains roughly 62-63 cents of gross profit per revenue dollar, with refunds running at only ~5% of revenue, which is healthy for e-commerce retail and providing headroom for promotional investment.
- Acquisition-channel concentration. Google search (gsearch) is the dominant and most efficient acquisition channel by both volume and revenue; Direct traffic, while smaller, converts best on a per-session basis. Socialbook underperforms on both volume and conversion.
- Heavy year-over-year concentration in 2014. Over 63% of both profit and revenue is attributable to 2014 alone, with a pronounced seasonal wave, mid-year and holiday-season(year end) peaks, spring/early-autumn troughs.
- New-customer risk & thin loyalty pipeline. First-time buyers carry materially higher churn/at-risk profiles than repeat buyers, and fewer than 1% of the user base progresses to the Gold loyalty tier despite a healthy Bronze-to-Silver transition.
- Under-leveraged cross-sell opportunity. Cross-sell conversion is strong in aggregate (roughly 1 additional item per 2 primary cart adds), but this lift is not evenly distributed to the lower-performing SKUs that would benefit most from it.

9.2 Recommendations

- Diversify the product portfolio's revenue mix, invest marketing and merchandising attention in The Forever Love Bear and The Birthday Sugar Panda to reduce single-SKU concentration risk.
- Protect and reinforce the gsearch channel with continued paid-search investment, while piloting incrementality tests on Direct-traffic drivers (email, retargeting) given their superior conversion rate.
- Design a first-purchase retention campaign (e.g. second-order discount, onboarding email sequence) targeted specifically at new customers to reduce the elevated at-risk share in this segment.
- Audit the Gold-tier loyalty threshold and benefits; a near-total absence of Gold members despite strong Bronze-to-Silver movement suggests the tier is either too hard to reach or insufficiently incentivised.
- Systematically surface cross-sell recommendations for The Hudson River Mini Bear and other lower-performing SKUs at the cart stage, where the data shows customers are already receptive to add-on suggestions.
- Use the 62%+ gross-margin cushion to fund targeted promotional pricing on off-peak months (March, September) to smooth the pronounced seasonal revenue trough.

10. Limitations & Future Scope

10.1 Limitations

- The product catalogue is limited to four SKUs, which constrains the depth of category-level or brand-level analysis possible in the current model.
- The dataset covers a fixed historical window (March 2012 to March 2015); no live or near-real-time refresh is configured, so the dashboard reflects a point-in-time snapshot rather than an operational, continuously updated system.
- Z-score-based outlier detection is a general-purpose statistical technique; with only four products, price and COGS naturally cluster into discrete values, which can inflate apparent outlier counts and requires analyst judgement to interpret correctly.

10.2 Future Scope

- Extend the semantic model with a customer lifetime value (CLV) measure and cohort-based retention curves to deepen the loyalty-tier analysis in Section 9.
- Introduce a scheduled refresh (e.g. via Power BI Service gateway) so the dashboard can operate against a live SQL Server connection rather than a static extract.
- Expand the product catalogue analysis with category/brand hierarchies once additional SKUs are onboarded, to reduce reliance on single-product concentration.

11. Conclusion

The Toy Store E-Commerce Sales Dashboard demonstrates a complete, disciplined analytics workflow, from raw, multi-source operational data through rigorous SQL Server data-quality assurance, a well-normalised star-style Power BI semantic model, a 36-measure DAX catalogue, and a seven-page interactive report suite bound together by consistent slicers and navigation.

The resulting dashboard gives The Toy Store's leadership team a single source of truth for revenue, profit, customer, product, transaction, inventory, and web-traffic performance, while the strategic takeaways captured on every page turn descriptive analytics into actionable business guidance. The core findings: strong margins, a dominant hero product, an efficient but concentrated acquisition channel, and a thin loyalty pipeline provide a clear, evidence-based agenda for the merchandising, marketing, and customer-retention initiatives outlined in the Key Findings & Business Recommendations section.

Delivered by an eleven-member analytics team over a two-month engagement, the project stands as a practical case study in translating raw e-commerce data into decision-ready business intelligence using the SQL Server and Power BI stack.

A. Appendix

A.1 Project Team

#	Team Member
1	Arpita Banik
2	Debanjan Sengupta
3	Puja Kumari
4	Rakhi Biswas
5	Riya Majhi
6	Rohit Ghosh
7	Sayan Ghosh
8	Siuli Bhowmick
9	Subham Hazra
10	Tanushree Seal
11	Tushar Mishra

A.2 Tools & Technologies

Tool	Role in Project
SQL Server Management Studio (SSMS)	Data uploading & checks for nulls, duplicates, Z-score outliers & business-rule validation
Power BI Desktop	Data modelling, relationships, Date table, DAX measure, Dashboard design